Statistical Report

Determinants of Belgian Household Energy Consumption

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Contents

1 Background **2**

2 Data management **2**

2.1 Data Cleaning 2

2.2 Data Coding 2

3 Descriptive Statistics **3**

3.1 Categorical variables 3

3.2 Continuous variables 3

4 Inferential Statistics **4**

4.1 Regression model 4

4.2 Contrast analysis 5

5 Conclusions **5**

6 Appendix **6**

6.1 Summary tables for non-continuous varibales 6

6.2 Summary tables for continuous varibales 8

6.3 Summary plots for continuous varibales 9

6.4 Relationship between Annual Energy Consumption and non-continuous variables. . . 13

6.5 Regression Model 15
6.6 Directed acyclic graph (DAG) 15

References **20**

1 Background

The goal of "*Carbon Neutrality*" is gaining increasing momentum in the European political arena amid substantial indicators of an incoming climate change which might have deleterious effects at the societal, economic and, most importantly, environmental level. Household energy consumption plays a pivotal role in overall social sustainability since it accounts for a substantial share of the total energy demand in most developed countries. Within this sector, electricity consumption provides a useful proxy for overall household energy demand and can be put into relationship to structural characteristics of the buildings as well as the socio-economic profile of their occupants.

Dwelling related attributes such as floor area, dwelling type, construction period, and Energy Performance Certificate (EPC) labels are consistently identified as important predictors of electricity demand (McLoughlin et al., 2013; Santin et al., 2009). Intuitively, larger dwellings typically require more energy for lighting, heating, cooling, and appliance use, while older buildings are constructed with materials which provides poorer insulation and therefore larger heat dissipation. Nevertheless, these variables interacts with socio-economic factors. Larger households topically host a higher number of occupants, therefore they can share the consumption for heating, refrigerators and air conditioning lowering the *per capita* consumption. Similarly, high EPC label buildings are prerogative of high-income classes which might use a larger number of appliances and indulge into energy-consuming behaviours going to erode the benefits of a properly insulated house in terms of energy savings (Aydin et al., 2017). This cross-sectional study collected data from 2500 households in Belgium and examines the relationship between households electricity consumption and several predictors.

2 Data management

2.1 Data Cleaning

Three observations (rows 66, 36, and 135) were removed from the dataset according to the instructions. Subsequently, the dataset was cleaned by excluding observations with missing values in the variables EPC label (`epc_label`) and annual household electricity consumption (`annual_kwh`). This step ensures that the analysis is conducted only on complete cases for the key variables of interest. Missing values in other variables were retained (see appendix) without imputation. None of the variables displayed missingness above 5%. Five observations in the`annual_kwh` (row 267, 1330, 1570, 369, 12) variable displayed values two orders of magnitude larger than the others (172 to 312 times larger) and, therefore, were removed as were attributed to imputation error (see section 6.2 for details).

2.2 Data Coding

Variables were coded according to their type: continuous, count, ordinal, categorical, or binary. Table 1 summarizes their descriptions and coding schemes.

Table 1: Data Dictionary

Name	Description	Type_Coding
annual_kwh	yearly electricity use	Continuous (kWh/year)
sq_meters	indoor living area	Continuous (m^2)
epc_label	energy performance certificate	Ordinal: A, B, ..., F
building_era	construction period	Ordinal: ref = Pre-1970, 1970-2000, post-2000
income_euro	net household income	Continuous (€/year)
occupancy_count	permanent residents	Count: 1, 2, ..., n
dist_to_brussels	distance to Brussels center	Continuous (km)
roof_color	roof tile color	Categorical: red, black, other
pet_ownership	owner has a pet	Binary: 0 = No, 1 = Yes

3 Descriptive Statistics

3.1 Categorical variables

The distribution of EPC labels is relatively balanced across categories. Labels A (22.2%) and C (21.3%) are the most common, followed by B (17.6%) and D (15.7%). Lower-efficiency categories E occur less frequently, 13.6% and F 9.6% respectively. Buildings constructed between 1970 and 2000 represent 34.4% of the sample, pre-1970 33.1%, and post-2000 32.6%. Occupancy count ranges from 1 to 7 with all counts accounting approximately 14-15%. A breakdown is provided in the appendix. None of the categories appear too sparse, suggesting that all variables can be retained for modelling without concerns about low cell counts. From the exploratory analysis, two categorical variables seem to correlate strongly with electricity consumption: EPC label and occupancy count. The other categorical variables do not display an apparent correlation with electricity consumption. Moreover, EPC label and building era are correlated with each other since the most efficient labels are clustered in the two most recent building eras. (see section 6.4).

3.2 Continuous variables

On average, households consume 5,874.76 kWh of electricity annually (outliers removed), with a relatively large standard deviation (2,224.25 kWh). The median consumption (5,432.07 kWh) is slightly lower than the mean, suggesting a right-skewed distribution, which is also supported by the QQ plot showing upward deviations above the diagonal line in the upper tail (see appendix). Floor area averages 158.25 m^2 with a standard deviation of 36.50 m^2, and a range from 60.26 m^2 to 341.30 m^2. Household income averages €48,842.90, with a standard deviation of €22,534.97 and a range from €8,487.91 to €206,367.91. As expected for this type of variable the QQ plot shows strong upper-tail deviation, suggesting positive skewness driven by high-income outliers. A breakdown for each variable is provided in section 6.2.

The correlation matrix reveals that annual electricity consumption has a moderate positive correlation with floor area (r = 0.38) and with income (r = 0.29), suggesting that larger and wealthier households tend to consume more energy. Importantly, two predictors exhibit a strong positive

correlation with each other, namely floor area and income (r = 0.72). Distance to Brussels shows virtually no correlation with the other variables, suggesting that location relative to Brussels does not strongly influence household energy use, dwelling size, or income in this dataset.

4 Inferential Statistics

4.1 Regression model

The initial model contained all variables plus interaction terms between income and number of occupants, EPC label and floor area, EPC label and number of occupants, and income and EPC label. Interactions were selected based on literature and exploratory analysis. A quadratic term was considered for `occupancy_count` consistently with a possible decreasing marginal effect. Predictor retention was determined using a backward stepwise procedure based on the Bayesian Information Criterion (BIC), which favours more parsimonious models. At each step, the variable whose removal produced the greatest reduction in BIC was eliminated, and the process continued until no further decrease was observed. The outcome variable `annual_kwh` was initially regressed using row data. Diagnostic plots (see appendix) showed some violations of the standard assumptions. In particular, the residuals vs fitted plot indicates the presence of heteroskedasticity, with the variance of the residuals increasing for higher fitted values, and mild deviations from normality in the Q-Q plot. The outcome variable `annual_kwh` was then log transformed resulting an an improvement of the diagnostic plots, particularly by yielding a more symmetric residual distribution and enhancing homoskedasticity (see section 6.5).

The final model retained floor area, EPC label, and the number of occupants as predictors of annual electricity consumption. The interaction terms and other covariates were removed. Nevertheless, `building_era` was retained in the model as it acts as a confounded being related to both `annual_kwh` and `epc_label` (see section 6.4 for details).

The results indicate that household size is a strong predictor of electricity consumption, with each additional square meter associated with an approximate 0.38% increase in annual consumption. Compared to dwelling in energy class A, lower efficiency classes exhibit progressively higher electricity consumption, from about 7% for class B to nearly 37% for class F. Each additional occupant is responsible for around 15-16% increase in electricity consumption. No evidence a decreasing marginal effect of the number of occupants was found at 5% significance level. The final model achieved a substantially lower BIC while maintaining nearly identical R^2 (89%), indicating that almost all the variance in annual consumption is explained by EPC label, house size and number of occupants.

Table 2: Regression results for the selected model

	Estimate	Std. Error	t value	p-value
(Intercept)	7.21982	0.02009	359.33464	0.00000
sq_meters	0.00396	0.00007	58.42909	0.00000
epc_labelB	0.08704	0.00929	9.36850	0.00000
epc_labelC	0.15666	0.01420	11.03117	0.00000
epc_labelD	0.23751	0.01491	15.93255	0.00000
epc_labelE	0.31590	0.01742	18.12943	0.00000

	Estimate	Std. Error	t value	p-value
epc_labelF	0.39357	0.01795	21.92164	0.00000
building_era1970-2000	-0.00860	0.00942	-0.91348	0.36108
building_eraPost-2000	-0.00954	0.01521	-0.62723	0.53057
occupancy_count	0.15110	0.00125	120.79605	0.00000

Table 3: Model comparison using AIC, BIC, and R-squared

	Model	AIC	BIC	R2
initial_model	initial_model	-3326.697	-3134.808	0.892
final_model	final_model	-3347.589	-3283.627	0.891

4.2 Contrast analysis

- for research question 2 and 4

5 Conclusions

6 Appendix

6.1 Summary tables for non-continuous varibales

Tables below refer to the dataset with only rows 66, 36 135 removed.

Table 4: EPC Label Distribution

epc_label	n	percent
A	554	22.2
B	436	17.5
C	532	21.3
D	391	15.7
E	338	13.5
F	241	9.7
NA	5	0.2

Table 5: Building Era Distribution

building_era	n	percent
1970-2000	856	34.3
Post-2000	814	32.6
Pre-1970	827	33.1

Table 6: Roof Color Distribution

roof_color	n	percent
Black	832	33.3
Other	788	31.6
Red	877	35.1

Table 7: Pet Ownership Distribution

pet_ownership	n	percent
0	1509	60.4
1	988	39.6

Table 8: Occupancy Count Distribution

occupancy_count	n	percent
1	352	14.1
2	357	14.3
3	346	13.9
4	379	15.2
5	377	15.1
6	342	13.7
7	339	13.6
NA	5	0.2

6.2 Summary tables for continuous varibales

Tables below refer to the dataset with only rows 66, 36 135 removed and the dataset after NA and outliers removed.

Table 9: Summary Statistics for Continuous Variables (row data)

Statistic	annual_kwh	sq_meters	dist_to_brussels	income_euro
Mean	14445.58	158.19	63.03	48844.80
SD	194579.20	36.47	33.07	22504.32
Median	5432.60	155.84	63.24	44557.61
IQR	2996.35	46.80	57.70	25871.05
Min	1824.63	60.26	5.11	8487.91
Max	5372190.00	341.30	120.00	206367.91

Table 10: Summary Statistics for Continuous Variables (cleaned dataset)

Statistic	annual_kwh	sq_meters	dist_to_brussels	income_euro
Mean	14445.58	158.19	63.03	48844.80
SD	194579.20	36.47	33.07	22504.32
Median	5432.60	155.84	63.24	44557.61
IQR	2996.35	46.80	57.70	25871.05
Min	1824.63	60.26	5.11	8487.91
Max	5372190.00	341.30	120.00	206367.91

6.3 Summary plots for continuous varibales

Figures below refer to the dataset used for the regression model from which missing values and outliers in annual energy consumption were removed as explained in section two. From the box plot we observe that several variables exhibit right-skewness and upper-tail outliers, particularly income and energy consumption. While annual energy consumption deviates significantly from normality, its log transformed analogue follows an approximate normal distribution dispute a mild deviation from the expected quantiles in the lower and upper tail. The QQ plot for floor area shows only mild departures from normality. The average distance to Brussels is 62.96 km. Its QQ plot indicates noticeable curvature, implying that the distribution deviates from normality, possibly reflecting clustering of households at certain distance ranges during the sampling procedure. Household income QQ plot shows strong upper-tail deviation, suggesting positive skewness driven by high-income outliers.

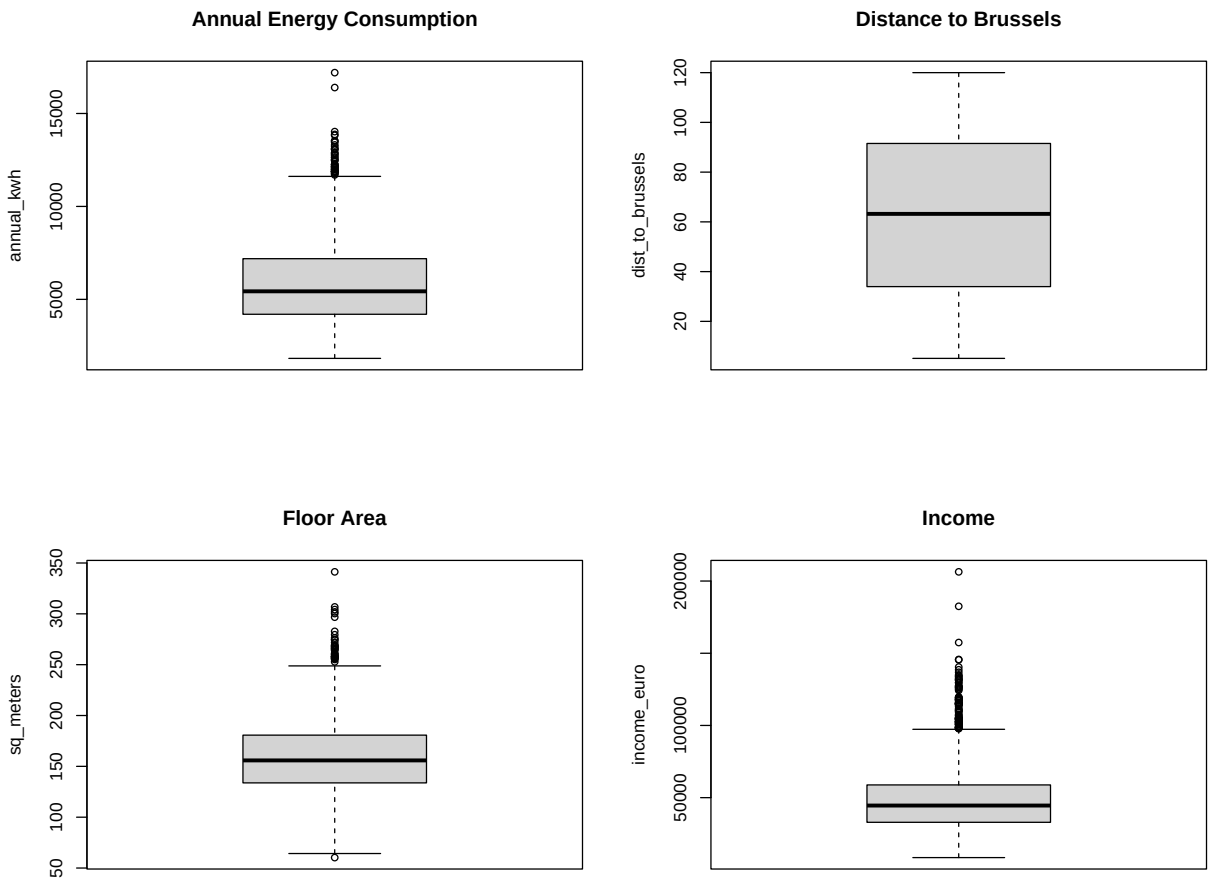


Figure 1: Box plots for continuous variables

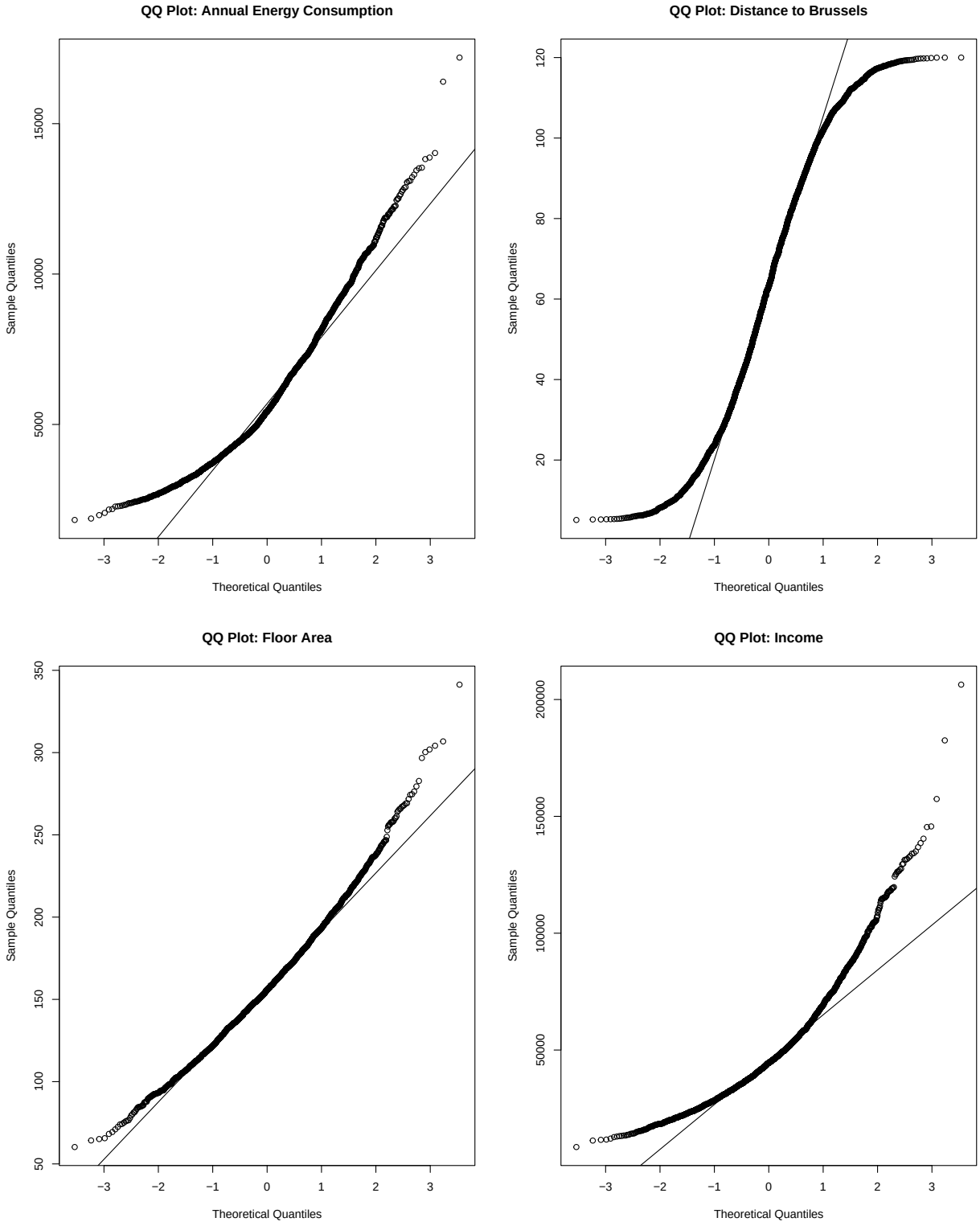


Figure 2: QQ plots for continuous variables

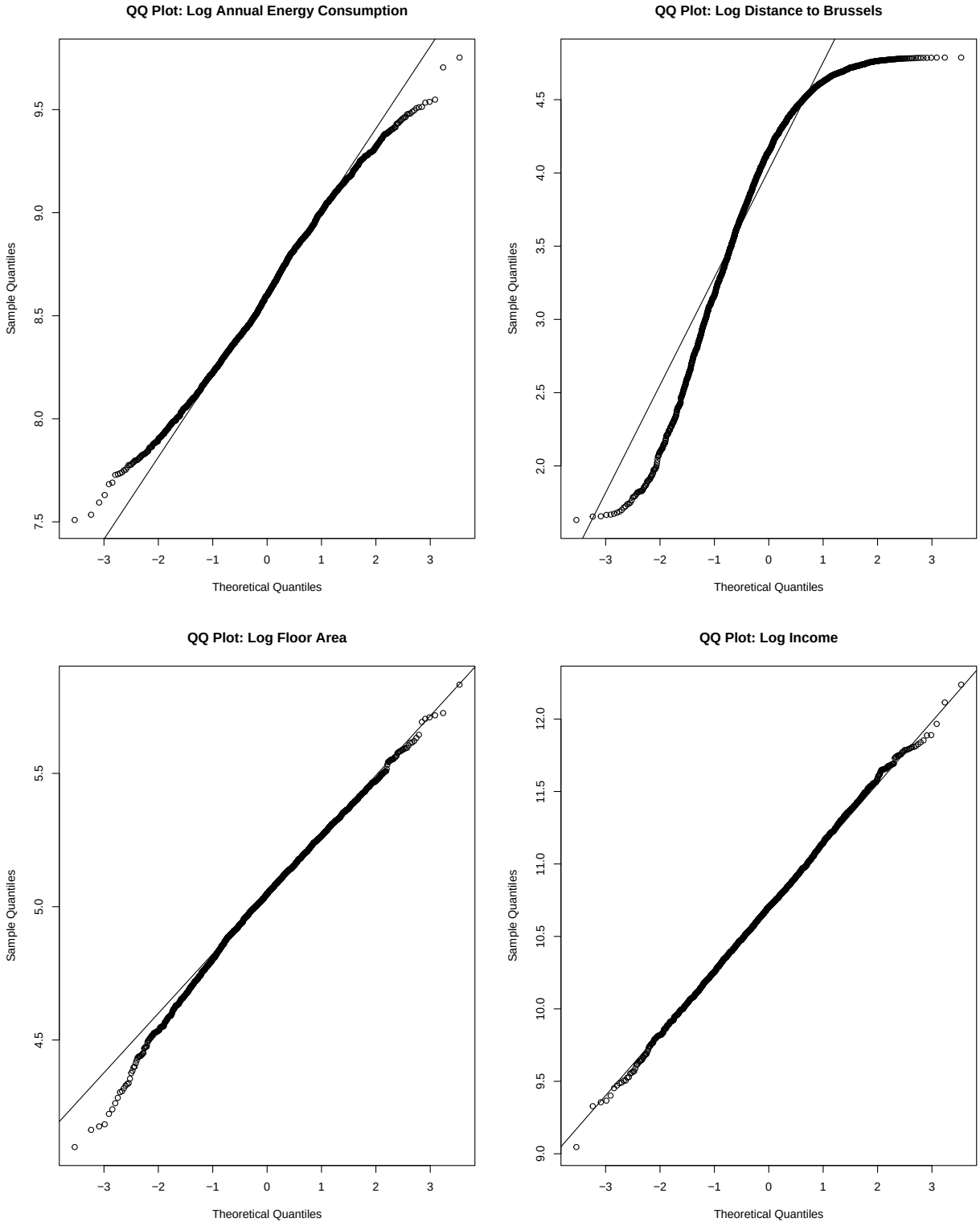


Figure 3: QQ plots for log-transformed continuous variables

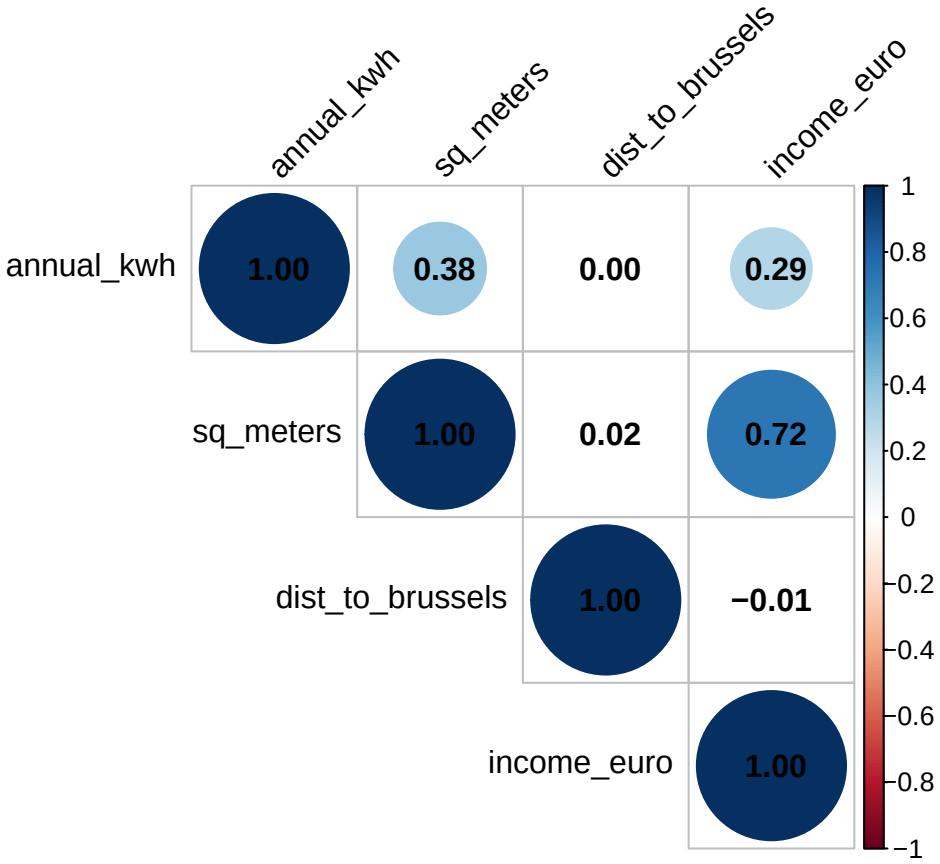
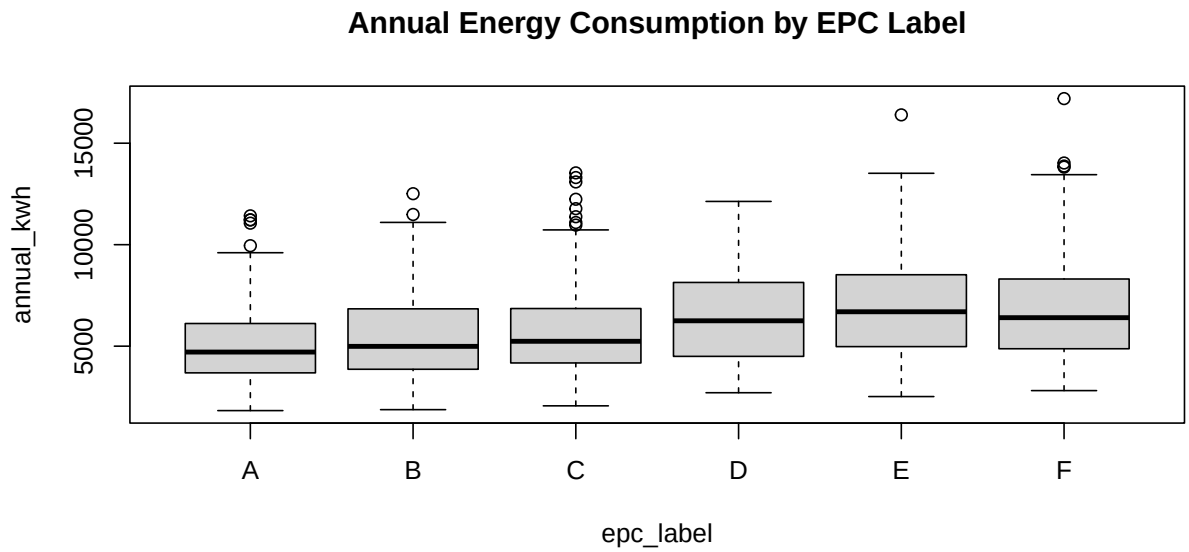


Figure 4: Correlation plots for continuous variables

6.4 Relationship between Annual Energy Consumption and non-continuous variables.

Among categorical and ordinal variables, EPC label and occupancy count are the most correlated to annual energy consumption. In addition, EPC label and building era are correlated with each other since the most efficient labels are clustered in the two most recent building eras and most recent buildings. Therefore, building era might play as a confounder in the model.



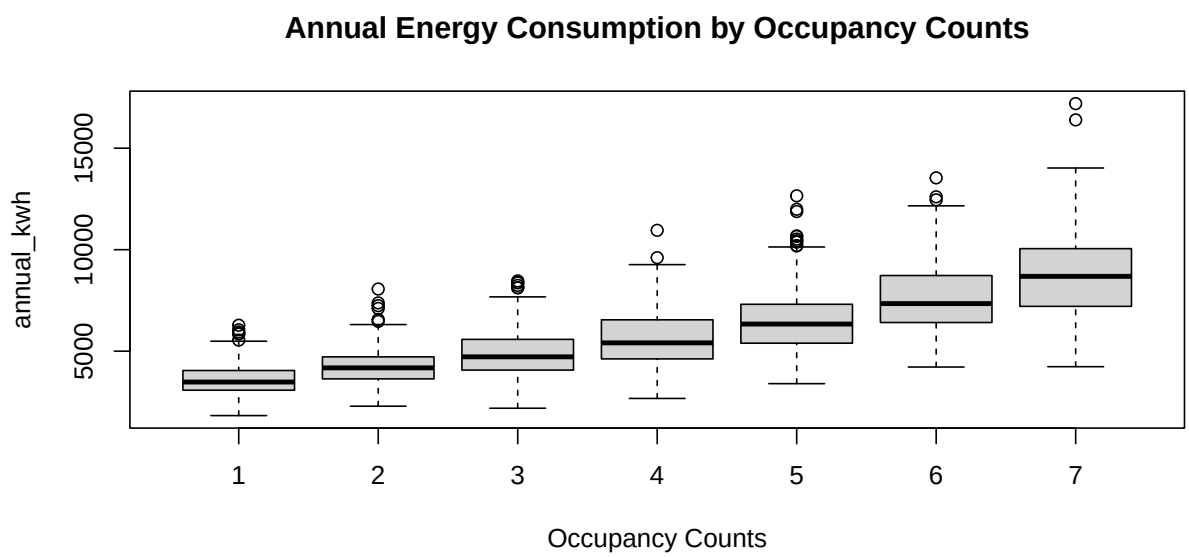
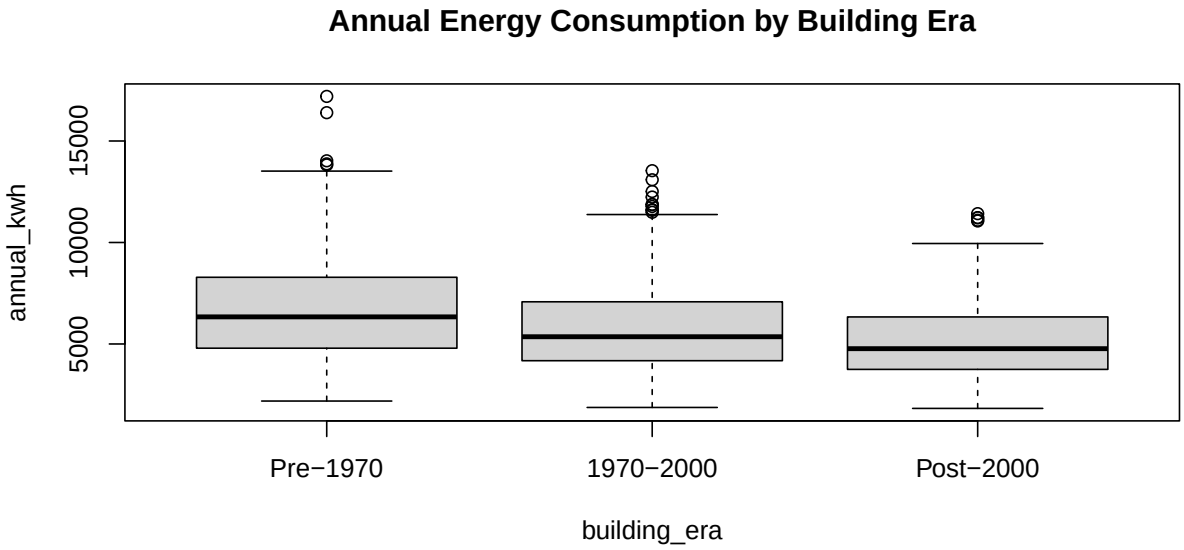


Figure 5: Box plots for annual energy consumption vs categorical variables (part 1)



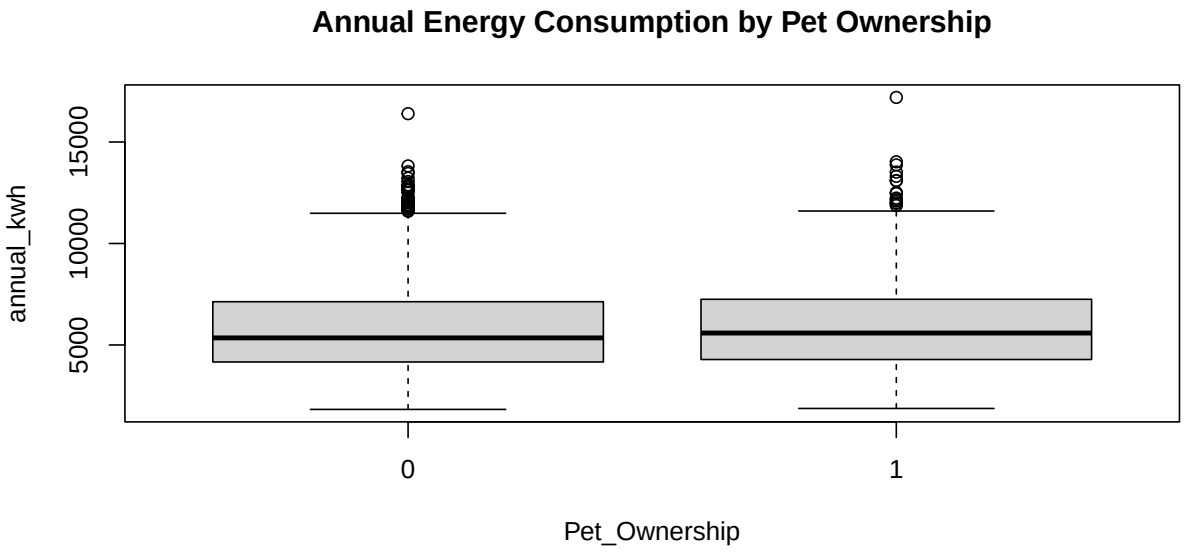


Figure 6: Box plots for annual energy consumption vs categorical variables (part 2)

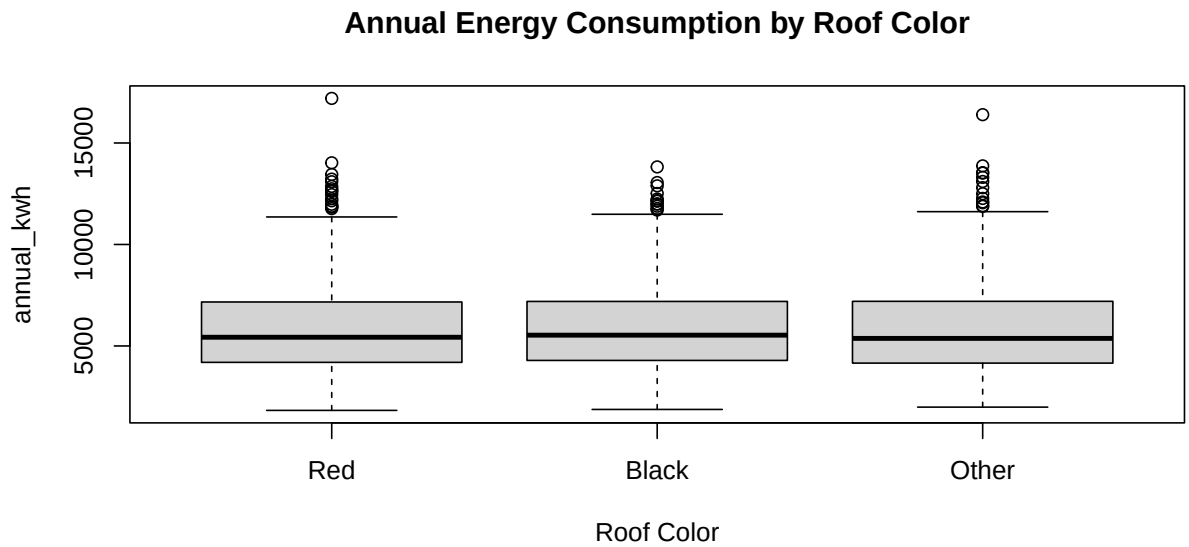


Figure 7: Box plots for annual energy consumption vs categorical variables (part 3)

6.5 Regression Model

The initial regression model was the following:

$$\begin{aligned}\text{annual_kwh}_i = \beta_0 &+ \beta_1\,\text{sq_meters}_i + \beta_2\,\text{income_euro}_i + \beta_3\,\text{occupancy_count}_i + \beta_4\,\text{dist_to_brussels}_i \\ &+ \boldsymbol{\beta}_5\,\text{epc_label}_i + \boldsymbol{\beta}_6\,\text{building_era}_i + \boldsymbol{\beta}_7\,\text{roof_color}_i + \beta_8\,\text{pet_ownership}_i \\ &+ \beta_9(\text{income_euro}_i \times \text{occupancy_count}_i) \\ &+ \boldsymbol{\beta}_{10}(\text{epc_label}_i \times \text{sq_meters}_i) \\ &+ \boldsymbol{\beta}_{11}(\text{epc_label}_i \times \text{occupancy_count}_i) \\ &+ \boldsymbol{\beta}_{12}(\text{epc_label}_i \times \text{income_euro}_i) \\ &+ \boldsymbol{\beta}_{13}(\text{occupancy_count}_i^2) + \varepsilon_i\end{aligned}$$

The final model was:

$$\log \text{annual_kwh}_i = \beta_0 + \beta_1\,\text{sq_meters}_i + \beta_2\,\text{occupancy_count}_i + \beta_3\,\text{epc_label}_i + \beta_4\,\text{building_era}_i + \varepsilon_i$$

6.6 Directed acyclic graph (DAG)

Building era was treated as a confounder in the relationship between EPC label and electricity consumption since it is related to both: the main predictor and the outcome variable.

Income was treated as a mediator in the relationship between EPC electricity consumption and size of the households since people with higher income can afford larger dwellings which in turn consume more electricity.

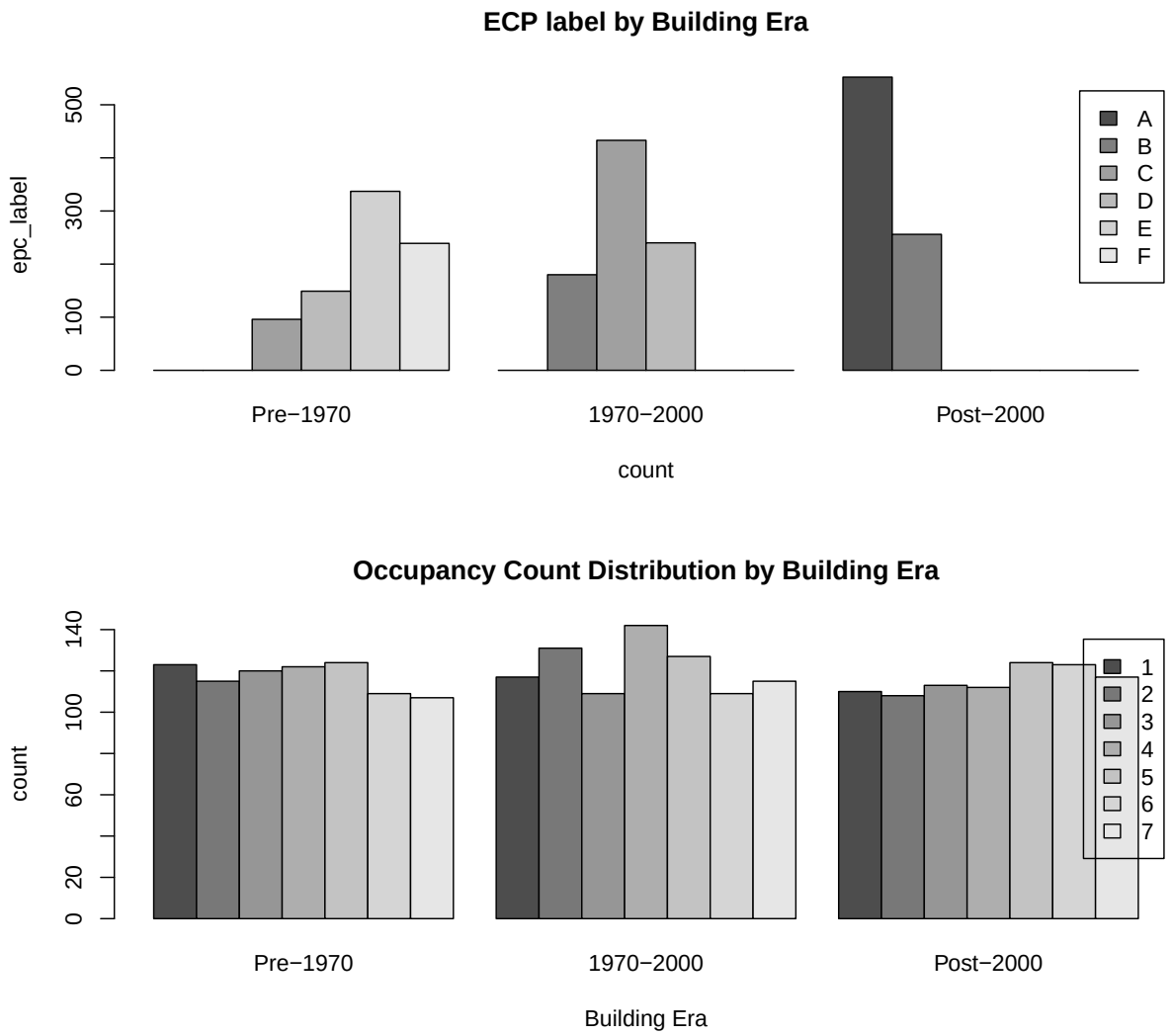


Figure 8: Bar plots predictors against each other (selected cases)

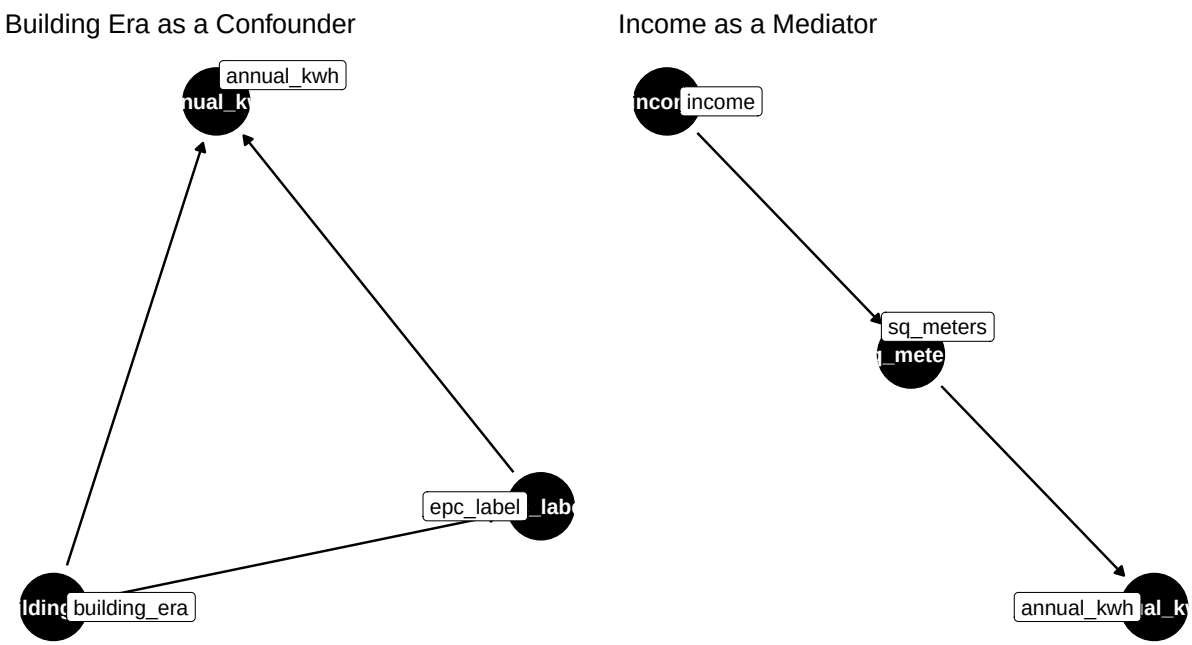


Figure 9: Directed acyclic graphs (DAGs)

6.6.1 Diagnostic plots: Model regressing row annual energy consumption

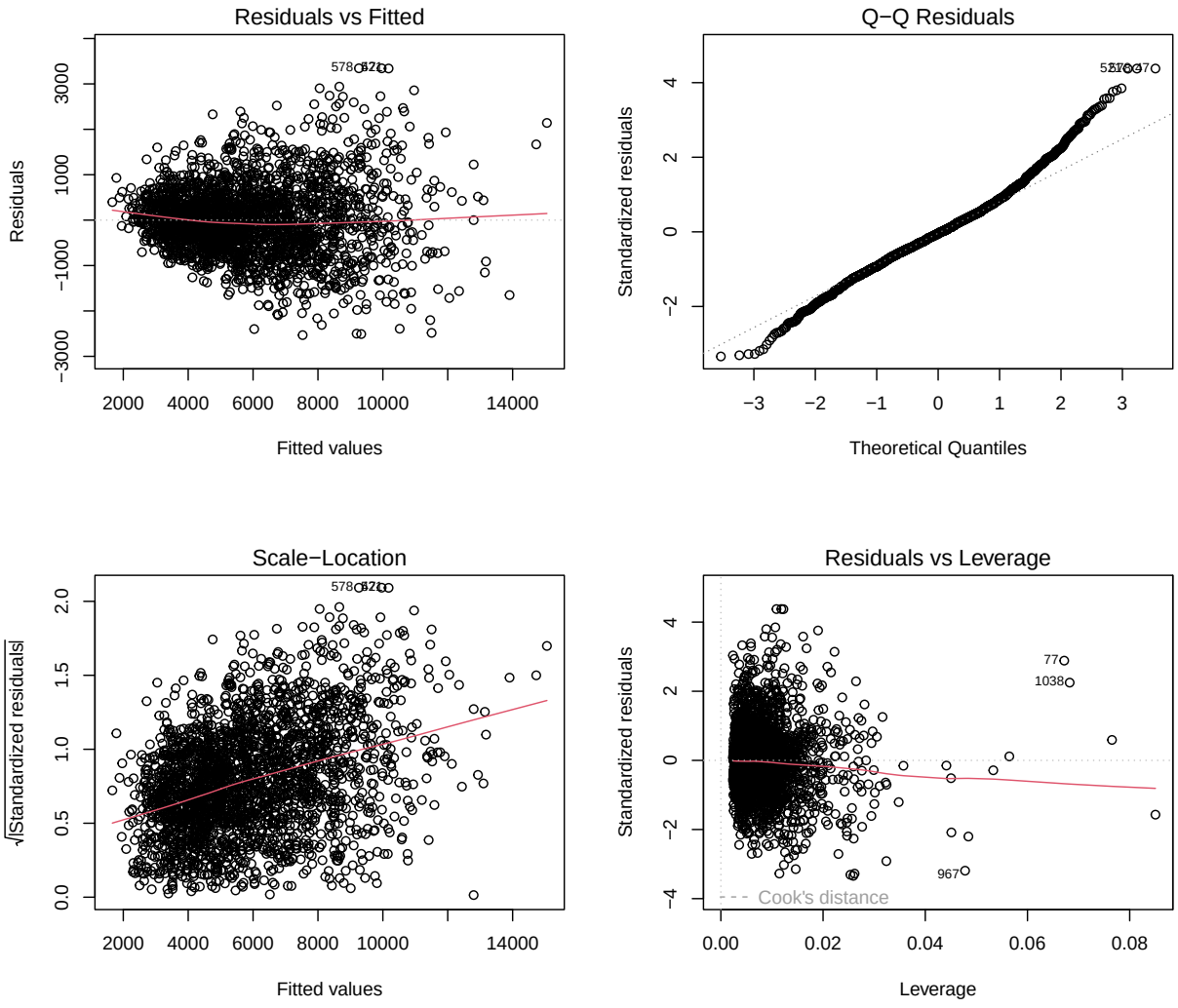


Figure 10: Diagnostic plots for the selected regression model using annual energy consuption without log transformation.

6.6.2 Diagnostic plots: Model regressing log-transformed annual energy consumption

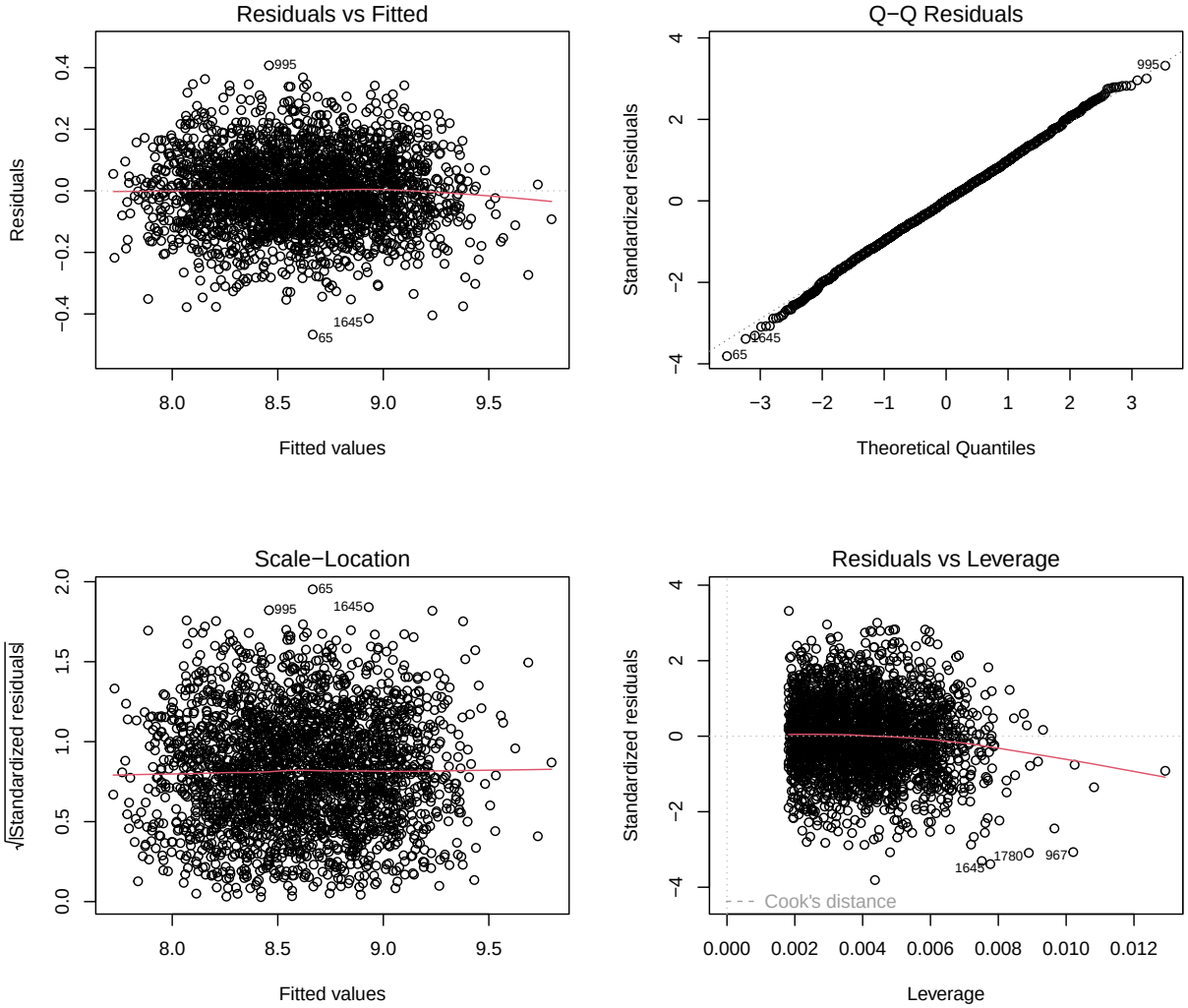


Figure 11: Diagnostic plots for the selected regression model using log-transformed annual energy consuption.

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